MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, SOCKET CONTACTS, 160 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS, (.100 INCH SPACING)

This specification is approved for use within Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

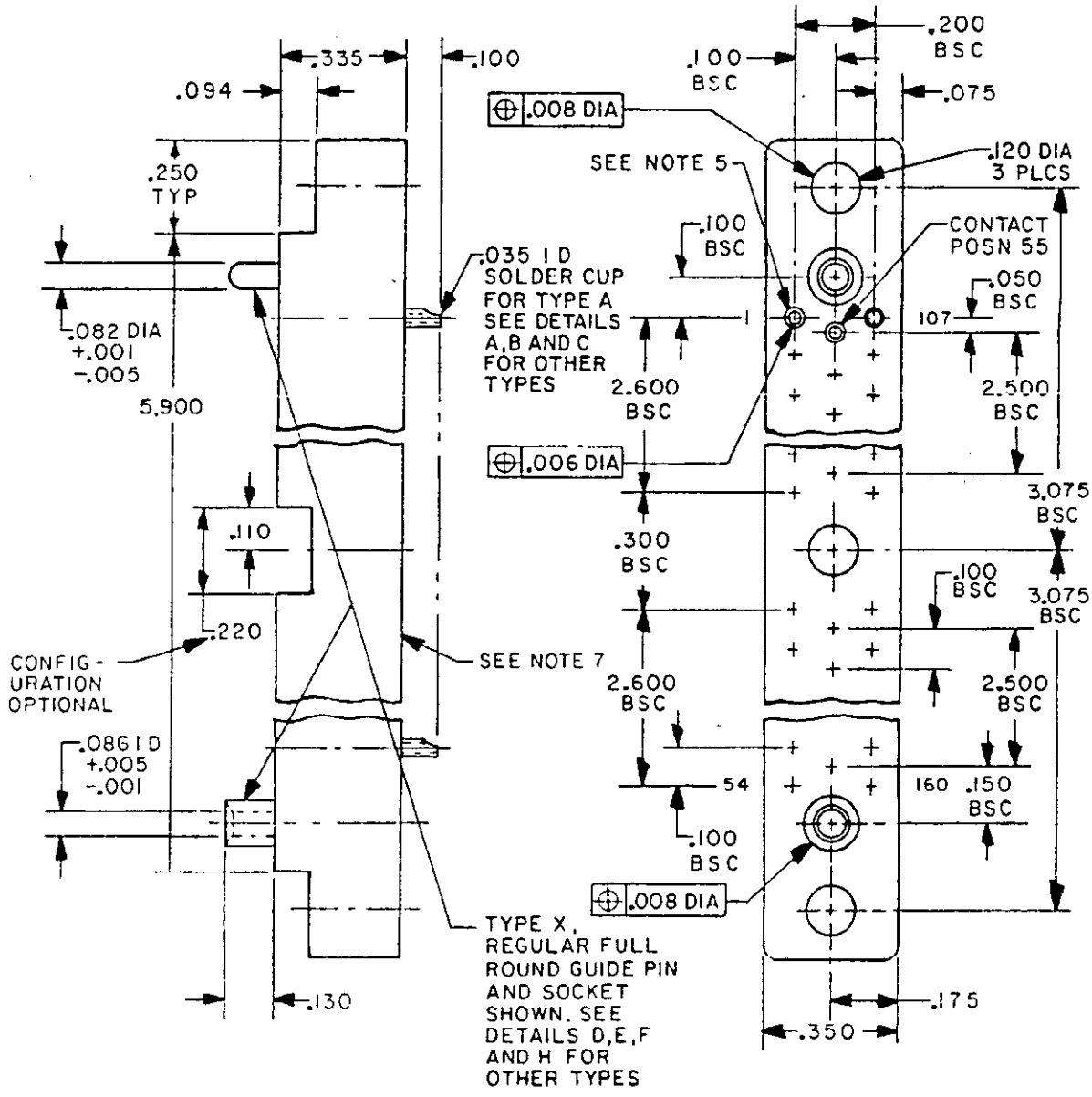
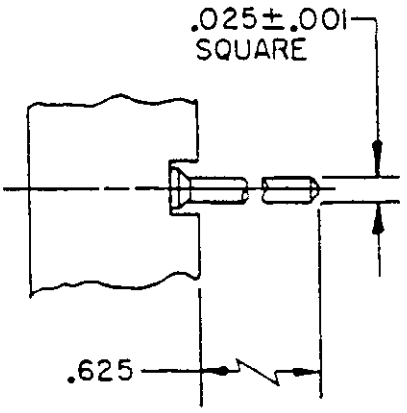


FIGURE 1. Connector, receptacle .100 inch (2.54 mm) spacing.

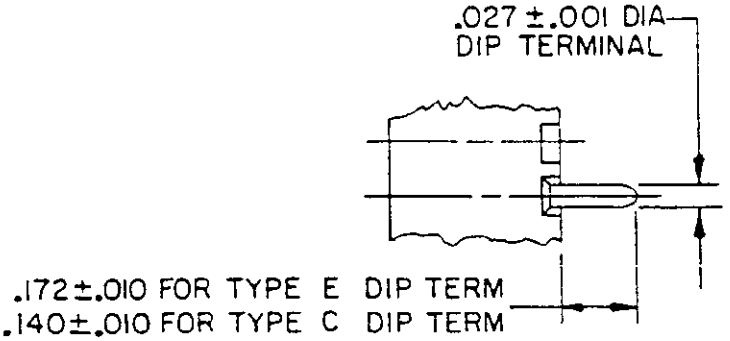
(A) denotes changes

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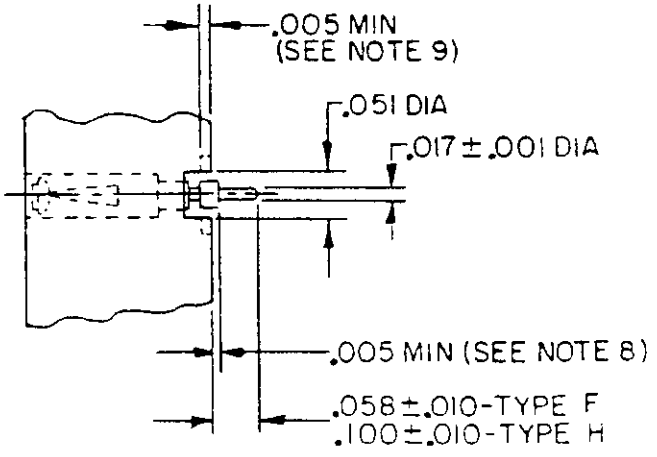


WIRE WRAP TERMINAL (TYPE D)

DETAIL A



DETAIL B

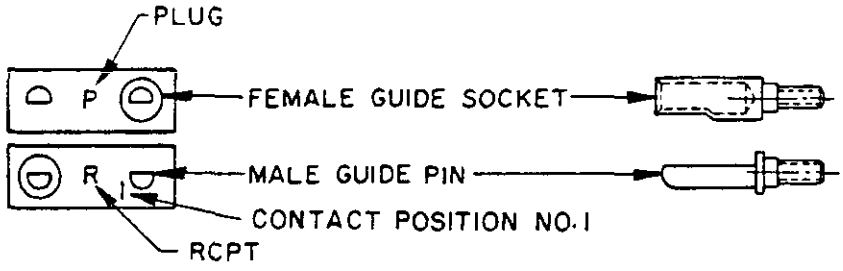


FLEXIBLE CIRCUIT TERMINAL

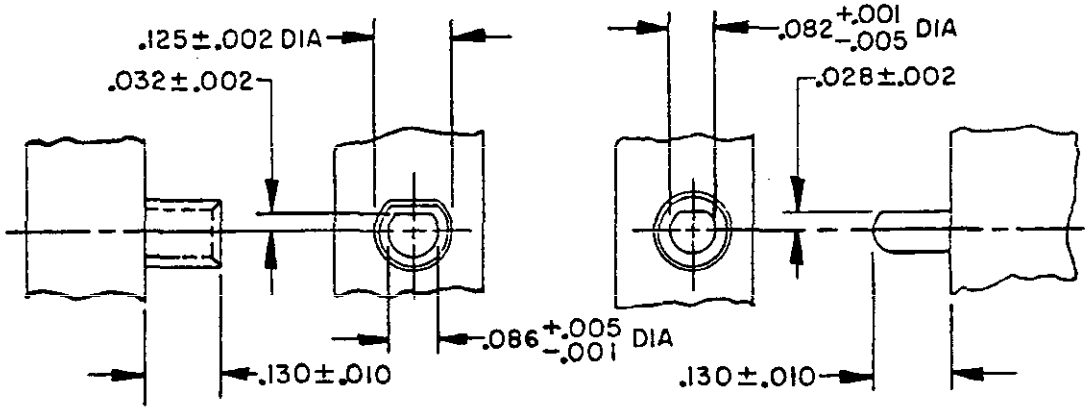
DETAIL C

FIGURE 1. Connector, receptacle .100 inch (2.54 mm) spacing - Continued.

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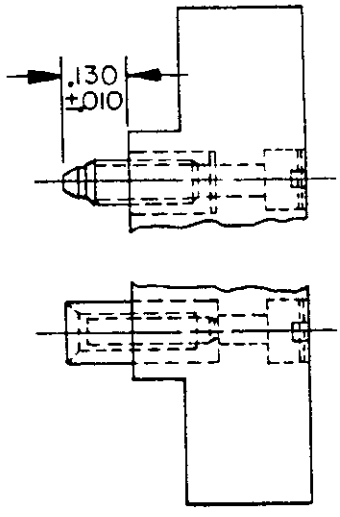
CONNECTOR POLARIZATION WITH D SHAPED GUIDE PINS AND SOCKETS. (TYPE Y)



DETAIL D

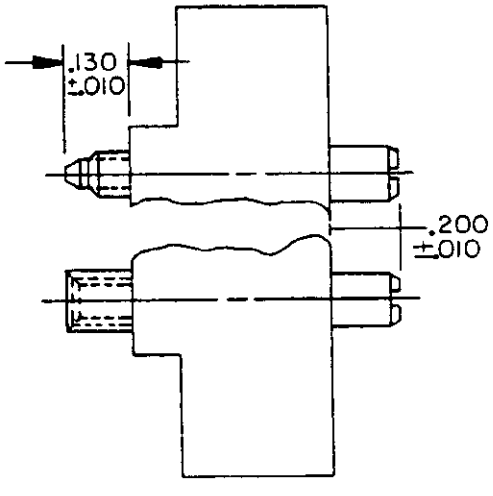
FIGURE 1. Connector, receptacle .100 inch (2.54 mm) spacing - Continued.

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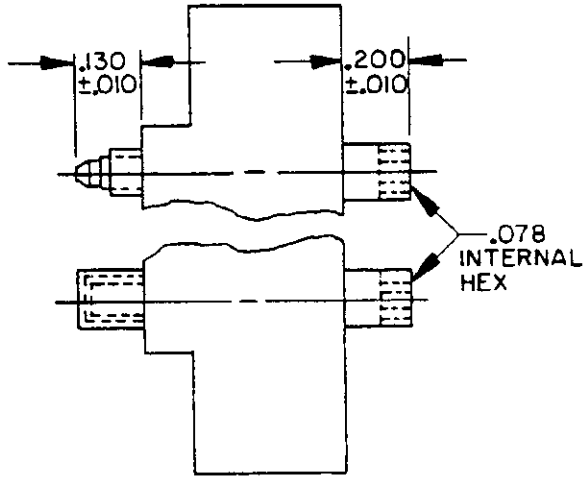
TYPE F FIXED JACKSCREW

DETAIL E



TYPE S SHORT TURNING SLOTTED JACKSCREW

DETAIL F



TYPE H SHORT TURNING SLOTTED JACKSCREW

DETAIL H

FIGURE 1. Connector, receptacle .100 inch (2.54 mm) spacing - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.027	0.69	.082	2.08	.130	3.30	.350	8.89
.002	0.05	.028	0.71	.086	2.18	.140	3.56	.625	15.88
.005	0.13	.032	0.81	.094	2.39	.150	3.81	2.500	63.50
.006	0.15	.050	1.27	.100	2.54	.172	4.37	2.600	66.04
.008	0.20	.051	1.30	.109	2.77	.175	4.44	3.075	78.10
.010	0.25	.058	1.47	.110	2.79	.200	5.08	5.900	149.86
.017	0.43	.075	1.90	.120	3.05	.220	5.59		
.025	0.64	.078	1.98	.125	3.18	.335	8.51		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise speicifed, tolerance is ±.005 (.13 mm).
4. These connectors mate with connectors specified in MIL-C-55302/138.
5. Numbers indicating the first and last contact on each row shall be stamped on the side. Every fourth contact shall be stamped on the first row only starting at the number one (1) contact with the exception that the 53 contact position may not be marked.
6. Termination layout on .025 (.64 mm) modular grid.
7. Optional .050 ±.010 (1.27 ±.25 mm) undercut to terminal side of connector for cleaning purposes.

FIGURE 1. Connector, receptacle .100 inch (2.54 mm) spacing - Continued.

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REQUIREMENTS

Dimensions and configuration: See figure 1.

Material: Guide pins, type X guide sockets, jackscrews and jack sockets shall be made of corrosion resisting stainless steel per ASTM-A582 passivated per QQ-P-35 or shall be corrosion resistant copper nickel alloy (61 ±2 percent copper, 25 ±2 percent nickel, 2.5 percent maximum other, and the balance zinc) and shall have a brinell hardness of 145 to 175. Type Y guide sockets shall be beryllium copper per QQ-C-550, nickel plated per QQ-N-290, class 2, 50 to 150 microinches.

(A) Plating: The contact plating shall be per MIL-C-55302, or gold per MIL-G-45204, type II, grade C, class 1 over nickel plating per QQ-N-290, 30 to 150 microinches in the engaging area and gold per MIL-G-45204, type II, grade C, class 00, over nickel plating per QQ-N-290, 30 to 150 microinches.

(A) Nonfunctional areas: See MIL-C-55302, paragraph 3.3.3.4.3, section C.

Contact identification: See figure 1.

Contact engagement and separation forces (No. 22 contacts):

Minimum separation: .5 ounce.
Maximum engagement: 12.0 ounces.

Mating and unmating: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.25 and the withdrawal force, in pounds, shall be a minimum of 0.025 times the number of contacts and shall not exceed the measured insert force.

Contact rating: 5.0 amperes maximum per contact.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .010 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Dielectric withstanding voltage:

Sea level: 1,000 volts rms.
High altitude: 300 volts rms.

Polarization: Type X, regular, guide pin and sockets, full round (see figure 1). Type Y, adjustable D-shaped guide pin and/or socket (see MIL-C-55302/57, detail B). All connectors are supplied in the -1 polarized position. When guides are removed and reinstalled in another position apply loctite 83-CV or equivalent to the threads. Ordering connectors in other than the -1 polarized position may be done by the original equipment manufacturer by placing the desired polarized position in brackets three spaces after the part number on the purchase order. Example: M55302/139-A100Y [-32]. Polarization pins and sockets shall be adjusted to any desired position by spanner wrench M55302/57-01. Ordering connectors in the -1 polarized position without loctite 83-CV or equivalent applied may be done by placing a "-0" in brackets three spaces after the part number on the purchase order.

Note: Hardware with Locktile 83-CV or equivalent applied shall withstand a torque of 7 in-oz applied through the nut.

Marking:

Connector part number: See example.

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Example of part number:

M55302/139 A 160 X

Basic number of specification sheet

Type of terminal (see figure 1)

A = Solder cup
B = Dip terminal .109 inch length
C = Dip terminal .140 inch length
D = Wrappost
E = Dip terminal .172 inch length
F = .058 flexible circuit terminal
H = .100 flexible circuit terminal

Number of contacts (see figure 1)

Type guide pin and/or socket (see figure 1)

X = Full round
Y = D-shaped
F = Fixed jackscrew
S = .200 slotted turning jackscrew
H = .200 hex turning jackscrew

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 85, 99

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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